

Part B-1**1. Excellence****1.1 Quality and pertinence of the project's research and innovation objectives****1.1.1 Introduction**

Continuous-variable (CV) quantum systems[1,2] are emerging as one of the leading platforms for quantum communication, sensing, networking, photonic quantum computing, and **Gaussian Boson Sampling (GBS)**[3-5]. However, their successful deployment increasingly depends not only on advances in quantum hardware but also on statistically reliable, physically consistent, and privacy-preserving data analysis. **Q-SHIELD** addresses this challenge by developing the first unified statistical framework for privacy-preserving inference in Hamiltonian-structured, energy-constrained bosonic quantum systems. By integrating rigorous statistical estimation, structure-aware statistical inference, hypothesis-driven experimental validation, and research software translation, the project establishes a complete methodological pipeline from statistical theory to experimentally validated and reusable computational tools. The proposed fellowship combines the applicant's expertise in statistical inference and **differential privacy (DP)** with Professor Morten Hjorth-Jensen's internationally recognised expertise in quantum modelling and computational physics, creating a unique interdisciplinary environment for developing trustworthy quantum information technologies. The project addresses the following scientific question: **How can physically meaningful information be inferred reliably from finite, noisy, energy-constrained quantum measurements under DP while preserving statistical efficiency and physical consistency?**

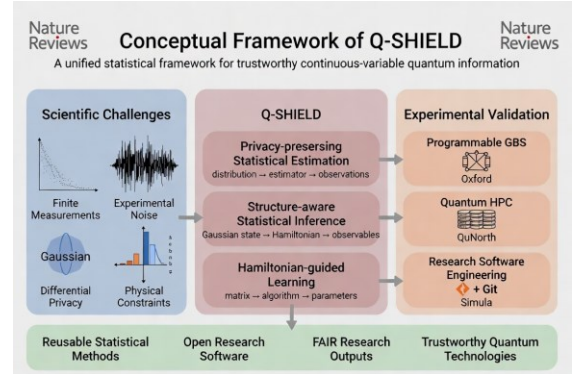


Figure 1. Conceptual framework of Q-SHIELD

1.1.2 Specific Objectives

The overall aim of **Q-SHIELD** is to establish a unified framework for privacy-preserving statistical inference in Hamiltonian-structured, energy-constrained bosonic quantum systems. To balance ambition with feasibility, the project adopts a staged strategy, establishing the core framework for analytically tractable Gaussian systems before extending it incrementally to controlled near-Gaussian systems [1,6], i.e., experimentally accessible perturbations of Gaussian reference models generated by weak nonlinear Hamiltonian corrections. Three closely connected research objectives, each aligned with a dedicated scientific work package, progressively quantify the statistical cost of privacy, exploit physical structure to retain efficient and physically consistent inference under privacy constraints, and validate these advantages under realistic experimental conditions.

O1. Privacy-preserving statistical estimation (WP1). Develop rigorous statistical estimation theory under DP for Hamiltonian-structured Gaussian and controlled near-Gaussian bosonic systems, using standard (ϵ, δ) -DP as the operational benchmark and a hypothesis-testing/f-DP perspective to derive finite-sample guarantees, minimax bounds, uncertainty quantification, and privacy-utility trade-offs. This objective will establish rigorous privacy guarantees and quantify the statistical cost of privacy under finite-sample and bounded-energy constraints.

O2. Structure-aware statistical inference under privacy constraints (WP2). Develop physically consistent statistical inference methods that exploit Hamiltonian structure, Gaussian-state models, sparsity, locality, symmetries, conservation laws, and bounded-energy constraints to retain statistical efficiency under the privacy constraints established in WP1. The quantum state will serve as the primary inference target, from which physically relevant quantities—including characteristic functions, Wigner representations, Hamiltonian parameters, statistical moments, and expectation values of observables—will be recovered. The objective will be verified through finite-sample statistical guarantees, privacy-utility performance, physical consistency, and uncertainty quantification [7,8].

O3. Hypothesis-driven experimental validation (WP3). Determine whether the privacy-utility advantages and physical reliability predicted by WP1–WP2 persist under realistic noise, loss, finite-sample, and privacy conditions. The framework will be validated on programmable GBS platforms and through scalable computational benchmarking by comparing structured with unstructured and private with non-private inference, followed by translation of the validated methodology into reusable research software during the non-academic placement. This objective will be

verified through experimental benchmarking, robustness evaluation, privacy–utility assessment, and reproducible software implementation.

1.1.3 State of the Art, Scientific Innovation and Ambition

Recent advances in CV quantum information, quantum statistical inference, programmable photonics, and **quantum differential privacy (QDP)** have established powerful tools for Gaussian-state reconstruction, Hamiltonian estimation, hypothesis testing, and privacy-preserving quantum information processing. However, these developments remain largely disconnected, leaving key challenges at the interface of privacy, statistical efficiency, physical modelling, and experimental validation unresolved.

SotA 1 – Privacy-preserving statistical estimation. Progress in Gaussian moment estimation, Hamiltonian parameter estimation, quantum hypothesis testing, and QDP has established connections between DP [9–11], quantum distinguishability, **quantum Fisher information (QFI)**, and quantum Cramér–Rao bounds. The f-DP framework represents privacy through hypothesis-testing trade-off functions, with **Gaussian differential privacy (GDP)** providing a tractable special family with tight composition properties [11]. Recent work has extended this viewpoint to (ϵ,δ) -QDP through quantum hypothesis testing and information ordering, yielding bounds on quantum f-divergences and SLD Fisher information [12]. However, an f-DP-based statistical inference theory for energy-constrained bosonic systems is still lacking, and the interaction of privacy with bosonic energy constraints, Hamiltonian structure, finite-sample estimation, and repeated measurements remains unresolved. **Q-SHIELD fills** this gap by establishing privacy guarantees and quantifying their statistical cost under Hamiltonian and bounded-energy constraints.

SotA 2 – Structure-aware statistical inference. Gaussian quantum-state tomography and recent Hamiltonian and structure-aware quantum learning methods have advanced state reconstruction and shown that physical structure can improve learning efficiency [1,2,13]. Yet Hamiltonian locality, sparsity, symmetries, conservation laws, and bounded-energy constraints are rarely integrated with privacy-preserving finite-sample inference. It therefore remains unclear whether physical structure can mitigate privacy-induced statistical degradation while maintaining physically admissible inference. **Q-SHIELD addresses** this gap by developing structure-aware quantum-state inference under the privacy constraints established in WP1, using physical structure to retain statistical efficiency and enforce positivity, normalization, and bounded-energy consistency. Characteristic functions, Wigner representations, Hamiltonian parameters, moments, and observable expectation values will then be recovered with uncertainty quantification [14,15].

SotA 3 – Experimental validation. Programmable GBS platforms, photonic quantum technologies, and large-scale quantum computing infrastructures increasingly enable experiments and benchmarking under realistic conditions [16,17]. Their use for rigorous validation of privacy-preserving statistical inference remains limited, particularly for assessing finite-sample performance, robustness, physical consistency, and privacy–utility trade-offs under noise and loss. **Q-SHIELD establishes** a hypothesis-driven validation framework comparing structured with unstructured and private with non-private inference to determine whether the predicted privacy–utility advantages and physical reliability persist under experimentally relevant conditions [18,19].

The scientific ambition of Q-SHIELD is to establish a new methodological paradigm for trustworthy CV quantum information processing by integrating rigorous statistical estimation, structure-aware statistical inference, hypothesis-driven experimental validation, and research software engineering into one coherent framework. Rather than advancing individual methodological components, the project develops a complete statistical methodology spanning theory, computation, experiment, and reproducible software.

The principal advances beyond the current state of the art are summarised in Table 1. Beyond its scientific contribution, the proposed framework provides reusable statistical methodologies applicable to quantum information science, photonic quantum technologies, quantum sensing, and future privacy-aware quantum computing, while supporting the strategic objectives of Horizon Europe in trustworthy and reproducible quantum technologies.

Table 1. Scientific advances beyond the current state of the art

Current state of the art	Q-SHIELD
Privacy and quantum statistical inference are studied largely independently.	Establishes a unified framework integrating DP with statistical estimation and inference for Hamiltonian-structured, energy-constrained bosonic quantum systems.

Current state of the art	Q-SHIELD
Existing quantum-state inference rarely exploits Hamiltonian structures, bounded-energy constraints, and physical priors within a unified statistical framework.	Develops physically consistent, structure-aware statistical inference that integrates Hamiltonian structures, Gaussian-state models, sparsity, locality, symmetries, and bounded-energy constraints.
Experimental quantum platforms are primarily used to demonstrate hardware performance or quantum advantage rather than statistically rigorous validation.	Introduces hypothesis-driven experimental validation with finite-sample guarantees, robustness analysis, uncertainty quantification, and privacy–utility evaluation under realistic experimental conditions.
Existing studies usually focus on isolated methodological components.	Establishes a complete methodological pipeline from statistical estimation, through structure-aware statistical inference, to experimental validation and subsequent translation into reusable research software.

1.2 Soundness of the proposed methodology

1.2.1 Overall Methodology

The proposed methodology has been designed to achieve the three research objectives through a coherent progression from statistical estimation, to structure-aware statistical inference, and finally to experimental validation. Each methodological stage provides the foundation for the next one. Theoretical results obtained in WP1 support the inference framework developed in WP2, while the methodologies established in WP2 are validated experimentally and computationally in WP3. This progression ensures that every objective is directly supported by an appropriate methodology and verified through clearly defined deliverables and milestones.

Scientific Assumptions. The methodology is based on three scientific assumptions. First, Hamiltonian-structured Gaussian bosonic systems provide an analytically tractable and experimentally relevant statistical model. Second, the energy constraint plays a fundamental statistical role: it defines the admissible class of bosonic states, controls otherwise unbounded observables, and enters explicitly into finite-sample estimation and privacy–utility bounds. Hamiltonian sparsity and locality, together with symmetries and conservation laws where available, provide additional physical structure for improving statistical efficiency. Third, the framework is extended incrementally to controlled near-Gaussian systems[20], defined as perturbations of Gaussian reference models generated by weak low-order polynomial Hamiltonian corrections. This restricted class provides a controlled route beyond Gaussianity.

Privacy Model and Statistical Representation. Q-SHIELD will use standard (ϵ, δ) -DP as the operational reference model, ensuring direct comparability with established QDP, while developing privacy–utility analysis through a hypothesis-testing and f-DP representation. At the operational level, neighbouring datasets will primarily differ in one measurement record or experimental run, thereby protecting the contribution of an individual quantum measurement to the released statistical analysis. At the theoretical level, the resulting privatized outputs will be represented by pairs of quantum states whose distinguishability is constrained through quantum hypothesis testing. This separation connects record-level privacy to state-level information bounds without conflating privacy of measurement data with privacy of the underlying physical state. Within the f-DP framework, particular attention will be given to GDP, whose Gaussian trade-off function provides a tractable model for privacy composition across repeated measurements.

Methodological Strategy. The methodology follows three consecutive stages. WP1 establishes non-asymptotic privacy-preserving estimation theory for first- and second-order moments, selected Hamiltonian parameters under joint energy and privacy constraints[21]. Hypothesis-testing representations of privacy will be connected to quantum distinguishability, SLD Fisher information, and quantum Cramér–Rao bounds to derive minimax rates, confidence bounds, and privacy–utility limits, while composition across repeated measurements will be analysed within the f-DP/GDP framework. WP2 converts these theoretical results into structure-aware quantum-state inference by exploiting Hamiltonian sparsity and locality, Gaussian reference models, and symmetries and conservation laws. Inferred states will satisfy positivity, normalization, and bounded-energy constraints; characteristic functions, moments, selected observables, and Wigner representations will then be recovered as derived quantities with uncertainty quantification. WP3 tests the resulting theoretical claims on programmable GBS data and through scalable computational benchmarking, comparing structured with unstructured and private with non-private baselines under realistic noise, loss, and finite-sample conditions.

Methodological Challenges. The principal methodological challenges are to obtain informative finite-sample privacy–utility bounds in infinite-dimensional bosonic systems, evaluate hypothesis-testing trade-off functions and their composition without excessive analytical complexity, maintain physical consistency under privacy perturbations, extend the framework incrementally beyond Gaussian reference models, and preserve theoretically predicted advantages under realistic experimental noise and loss. These challenges will be addressed through energy-constrained non-asymptotic analysis, comparison of f-DP/GDP and Rényi-based privacy accounting, Hamiltonian-guided modelling, physically constrained state inference, and hypothesis-driven experimental benchmarking.

Methodological Contribution of the Secondments. The planned secondments provide methodological capabilities that are essential for achieving the project objectives. The secondment at the University of Oxford, hosted by Professor **Ian A. Walmsley's** research group, provides access to programmable GBS platforms and advanced photonic quantum technologies required to validate the proposed statistical framework under realistic experimental conditions. This experimental capability directly supports WP3 and enables verification of the statistical guarantees established in WP1 and WP2. The secondment through QuNorth provides access to the Magne quantum computing infrastructure for large-scale computational benchmarking, allowing systematic evaluation of scalability, robustness, statistical accuracy, and privacy–utility trade-offs. Together, these complementary research environments ensure that the proposed methodology is both scientifically rigorous and experimentally verifiable.

1.2.2 Integration of Methods and Disciplines

Q-SHIELD integrates mathematical statistics with CV quantum information and statistical physics through a two-way methodological interaction. Finite-sample statistics, DP, and uncertainty quantification define the inferential guarantees and privacy–utility criteria for quantum-state and parameter estimation, while Hamiltonian structure, energy constraints, sparsity, locality, symmetries, and conservation laws constrain the admissible statistical models and provide physical information that can improve inference efficiency. This interaction is central to WP1–WP2: statistics informs privacy-preserving quantum inference, while quantum structure informs statistical efficiency and physical admissibility under privacy constraints. WP3 closes the theory–experiment loop by using programmable photonic platforms and large-scale computation to test whether these statistical and structural advantages persist under realistic noise, loss, and finite samples. Research software engineering subsequently translates the validated methodology into reproducible and reusable computational tools. Together, these interactions integrate statistical guarantees, physical structure, experimental validation, and computational implementation in a way that no single discipline can provide.

1.2.3 Gender Dimension and Diversity Aspects

The proposed research develops statistical methodologies for Hamiltonian-structured CV quantum systems and does not involve human participants, personal data, biological materials, or research questions where sex, gender, or other diversity characteristics influence the scientific objectives, methodology, or expected outcomes. Consequently, the gender dimension is not relevant to the scientific content of the proposed research. This assessment has been made following the Horizon Europe guidance on the integration of the gender dimension into research and innovation. Nevertheless, all research activities will be conducted in accordance with the UiO's policies on equality, diversity, inclusion, research integrity, and responsible research conduct.

1.2.4 Open Science Practices

Open Science is integrated throughout the proposed methodology to enhance transparency, reproducibility, methodological robustness, and long-term scientific reuse. Open dissemination of algorithms, computational workflows, research software, and benchmark datasets will facilitate independent verification, comparative evaluation, and future methodological development.

In accordance with Horizon Europe requirements, all peer-reviewed publications will be made immediately available through trusted open-access repositories. Statistical inference algorithms, reusable research software, simulation codes, benchmarking workflows, software documentation, and computational protocols will be released under appropriate open-source licences whenever possible. The research software developed during the non-academic placement will be released together with documentation, benchmarking workflows, and reproducible computational pipelines whenever possible. These resources will enable other researchers to reproduce, validate, and extend the proposed methodology.

Research data and other research outputs will be managed according to the FAIR principles (Findable, Accessible, Interoperable and Reusable). A **Data Management Plan (DMP)** will be prepared during the early stage of the fellowship and updated throughout the project. Simulation datasets, processed experimental data, benchmarking results, reproducible computational workflows, and associated software will be deposited in trusted public repositories whenever possible, unless restricted by legal, ethical, or intellectual property considerations.

1.3 *Quality of the supervision, training and of the two-way transfer of knowledge between the researcher and the host*

1.3.1 Qualifications and Experience of the Supervisors

Professor Morten Hjorth-Jensen provides highly complementary expertise for Q-SHIELD in computational physics, quantum many-body theory, machine learning, and quantum information. His recent work on **Parametric Matrix Models (PMMs)**, physics-informed machine learning, and quantum computing directly supports the Hamiltonian-guided modelling and structure-aware inference developed in WP2, complementing the applicant's expertise in statistical inference and DP. His leadership of the proposed **Centre for Quantum Control (QONTROL)** further provides an interdisciplinary environment connecting theoretical modelling, advanced computation, and quantum information.

Professor Hjorth-Jensen has extensive experience leading international research projects funded by the European Union, the Research Council of Norway, the U.S. National Science Foundation, and the U.S. Department of Energy. He has supervised more than 100 Master's students, over 30 PhD candidates, and numerous postdoctoral researchers, many of whom have progressed to independent academic careers. This combination of scientific leadership, international collaboration, and advanced supervision provides a strong foundation for both the implementation of Q-SHIELD and the fellow's development towards research independence.

The supervisory environment is complemented by **Professor Ian A. Walmsley's** research group at the University of Oxford and **Dr. Shaukat Ali** at SIMULA Research Laboratory. Their expertise in programmable photonic quantum technologies and research software engineering complements the theoretical and statistical expertise at UiO, supporting experimental validation in WP3 and the subsequent translation of validated methods into reproducible research software.

1.3.2 Planned Training Activities for the Researcher

The training programme is tailored to complement the applicant's established expertise in statistical inference, DP, information theory, and statistical learning with the competences required to extend this research towards realistic CV quantum systems. The principal scientific training needs are Hamiltonian-guided quantum modelling, CV quantum information, experimental photonic technologies, large-scale quantum benchmarking, and research software engineering. These competences will be acquired through training-through-research under Professor Morten Hjorth-Jensen's supervision and within QONTROL, complemented by targeted training at Oxford, QuNorth, and SIMULA, directly supporting WP2–WP3.

To support the transition to independent research leadership, complementary transferable competences in project management, grant writing, Open Science and **research data management (RDM)**, scientific communication, supervision, and career planning will be developed through project implementation, UiO training, THE WEEK, individual career counselling, co-supervision, and scientific dissemination. The mandatory **Career Development Plan (CDP)** will be established by M6 and reviewed during the fellowship to align training with evolving research and career needs. The main training activities and their contribution to Q-SHIELD are summarised in Table 2.

Table 2. Targeted training programme and contribution to Q-SHIELD

Training activity	Provider	Contribution
Quantum Computing and Quantum Machine Learning (FYS5419/9419)	Prof. Morten Hjorth-Jensen	Quantum computing and Hamiltonian-guided modelling (WP2)
Advanced Machine Learning (FYS5429/9429)	Prof. Morten Hjorth-Jensen	Structure-aware statistical learning and scalable computation (WP1–WP2)
QONTROL supervision, seminars and journal clubs	QONTROL / Prof. Hjorth-Jensen	CV quantum information, interdisciplinary modelling and scientific exchange (WP2)
Programmable photonic quantum technologies	Oxford	Experimental CV/GBS methods and validation (WP3)
Large-scale quantum benchmarking	QuNorth / Magne	Scalable benchmarking and robustness analysis (WP3)
Research software engineering and reproducibility	SIMULA	Reusable software, benchmarking workflows and Open Science (WP3/NAP)
Grant writing, project management, Open Science/RDM, THE WEEK and career counselling	UiO	Research leadership, funding competence, career planning and transferable skills

1.3.3 Two-way Transfer of Knowledge

Knowledge transfer from the applicant to the host. The applicant contributes extensive expertise in statistical inference, probability theory, information theory, DP, **expectation-maximisation (EM)** algorithms, and structured statistical learning. Previous research on privacy-preserving density estimation, federated learning, Pufferfish privacy, and QDP provides methodological foundations that are currently complementary to the host's research programme. These competences will strengthen ongoing research at the UiO by introducing rigorous finite-sample statistical analysis, uncertainty quantification, statistical guarantees, and privacy-aware inference into Hamiltonian-structured quantum systems.

Knowledge transfer from the host to the applicant. The UiO will provide complementary expertise in structured quantum modelling, Hamiltonian-guided computational methods, CV quantum information, quantum many-body theory, programmable photonic quantum technologies, and high-performance scientific computing. These capabilities will enable the applicant to extend established statistical methodologies towards realistic quantum systems while developing expertise in experimentally relevant quantum information processing. The planned secondments and the non-academic placement will further broaden the applicant's experience through experimental validation, computational benchmarking, and research software engineering.

1.3.4 Rationale and Added Value of the Non-academic Placement and Secondments

The academic secondment at the University of Oxford, hosted by **Professor Ian A. Walmsley's research group**, provides complementary expertise in programmable GBS and photonic quantum technologies. It enables validation of the proposed statistical framework on state-of-the-art experimental platforms and broadens the fellow's expertise in continuous-variable quantum information research.

Complementary computational validation will be supported through **QuNorth's Magne** quantum computing infrastructure, providing access to large-scale quantum computing resources for benchmarking scalability, robustness, and privacy–utility trade-offs under realistic computational conditions.

The six-month non-academic placement at **SIMULA Research Laboratory**, supervised by Dr. Shaukat Ali, provides complementary training in research software engineering, benchmarking methodologies, reproducible computational workflows, and Open Science.

1.4 Quality and Appropriateness of the Researcher's Professional Experience, Competences and Skills

The applicant has developed a coherent research trajectory in statistical inference, information theory, and DP, progressively extending this expertise towards privacy-preserving quantum information. His published work provides direct methodological foundations for Q-SHIELD. In particular, his 2025 Scientific Reports study on Pufferfish privacy for Gaussian mixture models developed privacy mechanisms with information-theoretic guarantees, supporting the privacy-preserving estimation and privacy–utility analysis of O1/WP1 [22]. His broader work on DP density estimation, EM algorithms, and structured statistical learning [23–26] provides complementary expertise in statistical estimation, uncertainty analysis, and computational inference relevant to WP1–WP2.

This trajectory has developed further through the applicant's role as PI of the externally funded QDP for Irregular Data project (National Cybersecurity Authority, Saudi Arabia; €20.8k), providing direct research experience at the interface of DP and quantum information. Together with leadership of previous research projects, this demonstrates the capacity to formulate independent research questions, manage funded research, and develop an interdisciplinary programme connecting statistics, privacy, and quantum information.

Q-SHIELD builds on these established strengths while addressing complementary gaps in Hamiltonian-guided quantum modelling, CV quantum information, experimental photonic technologies, large-scale quantum benchmarking, and research software engineering through the expertise of UiO, Oxford, QuNorth, and SIMULA. This combination provides both the existing methodological foundation and the targeted new competences required to implement Q-SHIELD and establish an independent research direction in trustworthy statistical inference for quantum systems.

2. Impact

2.1 *Credibility of the measures to enhance the career perspectives and employability of the researcher and contribution to his/her skills development*

The applicant's long-term goal is to become an internationally competitive researcher leading an independent programme at the interface of mathematical statistics, privacy, and quantum information, with the capacity to establish a research group and compete for PI-level European funding. Building on his established expertise in statistical inference, DP, and project leadership, the next career stage requires advanced competences in quantum modelling, experimental CV technologies, large-scale computational benchmarking, research software engineering, and international research leadership. Q-SHIELD provides this progression through mentoring by Professor Morten Hjorth-Jensen and participation in QONTROL, experimental training at Oxford, large-scale benchmarking at QuNorth, and research software engineering at SIMULA. These complementary experiences will establish a theory–experiment–computation profile and an international network for leading interdisciplinary research in trustworthy quantum technologies.

The fellowship will further strengthen the transferable competences required for independent research leadership. Co-supervision opportunities will extend the applicant's substantial teaching and mentoring experience towards research supervision in an international quantum environment. Management of the fellowship's research and training resources will extend his existing project-management experience to European funding requirements. UiO research-funding activities will strengthen proposal-development skills and competitiveness for future ERC, Horizon Europe, and national PI-level funding, while THE WEEK and individual career counselling will support strategic career planning within and beyond academia. Training in scientific communication, Open Science, and project leadership will further strengthen research visibility and organisational capacity. Together, these measures will enhance competitiveness for independent academic positions, support the establishment of an international research group, and strengthen the capacity to lead externally funded research. The experimental, computational, and research-software competences gained through Oxford, QuNorth, and SIMULA will also broaden employability in quantum technology, scientific computing, and research-intensive software environments. The main career-development measures and expected benefits are summarised in Table 3.

Table 3. Career development measures

Career development measure	Competence developed	Expected career benefit
Mentoring by Prof. Hjorth-Jensen	Research strategy, scientific leadership, interdisciplinary collaboration	Leadership of an independent research programme
Participation in QONTROL	International networking and interdisciplinary collaboration	Sustainable international collaborations and research visibility
Oxford secondment	Experimental CV quantum technologies and theory–experiment integration	Capacity to lead interdisciplinary quantum research
QuNorth (Magne)	Large-scale quantum benchmarking and computational methods	Competitiveness for computational and quantum-technology research
SIMULA non-academic placement	Research software engineering, reproducibility and Open Science	Broader academic and research-intensive-sector employability
UiO research-funding activities	European and national proposal development	Competitiveness for independent research funding
THE WEEK and career counselling	Career planning, communication and transferable skills	Informed career development within and beyond academia
UiO internally funded additional year ¹	Teaching, supervision, exploitation of results and follow-on funding	Consolidation towards independent research leadership

2.2 Suitability and quality of the measures to maximise expected outcomes and impacts, as set out in the dissemination and exploitation plan, including communication activities

Dissemination and Exploitation. An initial dissemination, exploitation and communication (D&E&C) strategy will be implemented from M1 and updated alongside major project results. Q-SHIELD targets 4–6 Open Access publications in leading journals (e.g. PRX Quantum, Physical Review Letters, Quantum) and at least three presentations at major international conferences. Dissemination will follow project maturity: WP1 privacy and statistical results will be presented from M6–M12, WP2 structure-aware inference results from M15–M21, and WP3 experimental and computational validation results from M21 onwards. Target venues include QIP for theoretical quantum-information results, the European Quantum Technologies Conference (EQTC) for engagement with the European quantum research and innovation community, and IEEE Quantum Week for computational, engineering and software-oriented outcomes. WP1 outputs will primarily target researchers in quantum statistical inference, DP and quantum information theory; WP2 methods will target researchers in CV quantum information, Hamiltonian learning and computational quantum physics; and WP3 protocols, benchmarking resources and software will target experimental photonics groups, quantum-computing researchers and research-software developers. Publications and reusable research outputs will be released through NVA, GitHub and Zenodo, as appropriate, to facilitate validation, uptake and reuse.

Intellectual Property and Exploitation. Scientific reuse of the statistical methods, validation protocols and research software is the primary exploitation pathway. No patents are currently anticipated. Results with commercial potential will be assessed with UiO, INVEN2 and the UiO Growth House before disclosure to determine appropriate protection and exploitation measures while preserving Open Science principles.

Communication and Public Engagement. Communication will run from M1 throughout the project, centred on the message that trustworthy quantum technologies require privacy protection, statistically reliable inference and physical consistency. At least three public-engagement activities will target defined non-specialist audiences through European Researchers' Night, UiO Science Week/Open Day, and QONTROL/UiO outreach opportunities. UiO institutional communication channels will communicate major project results in accessible language. Progress will

¹ Subject to UiO eligibility rules; the additional year is internally funded and outside the MSCA action.

be monitored against targets of 4–6 Open Access publications, at least three major international conference presentations, at least three public-engagement activities, and one reusable software/reproducibility package. The principal communication activities are summarised in Table 4.

Table 4. Communication and public engagement plan

Activity	Target audience	Key message	Timing/KPI
European Researchers' Night	Students, young people & families	Privacy and reliable statistics for quantum technologies	≥1 activity
UiO Science Week / Open Day	Prospective students	Statistics and physics enable reliable quantum inference	≥1 activity
QONTROL / UiO outreach	Upper-secondary students	Q-SHIELD results and societal relevance	≥1 activity/output

2.3. The magnitude and importance of the project's contribution to the expected scientific, societal and economic impacts

Scientific impact. Q-SHIELD will establish a unified framework connecting privacy guarantees[21-23], statistical efficiency, physical consistency, and experimental validation in Hamiltonian-structured CV quantum systems, directly addressing the gaps identified in Section 1.1. The project will deliver rigorous statistical methods and finite-sample guarantees, structure-aware inference and validation protocols, 4–6 Open Access publications, and reusable research software. These outputs will be relevant across quantum statistical inference, DP, CV quantum information, Hamiltonian learning, computational quantum physics, and quantum machine learning. Their importance lies in enabling researchers to quantify the statistical cost of privacy, exploit physical structure to retain efficient and physically admissible inference, and test these advantages under realistic conditions. In the short term, open methods and benchmarking resources will support reproducibility and comparative evaluation; in the longer term, they can underpin new research on privacy-preserving quantum learning, Hamiltonian learning, and trustworthy quantum information processing.

Societal impact. Q-SHIELD contributes to the responsible development of quantum technologies by establishing methods that jointly address privacy protection, statistical reliability, and physical consistency under realistic noise and finite-data conditions. Its broader societal relevance lies in supporting trustworthy development of emerging quantum technologies and increasing understanding of privacy and reliable inference in quantum systems. At least three targeted activities through European Researchers' Night, UiO Science Week/Open Day, and QONTROL/UiO outreach will engage young people, students, families, and science-interested audiences. These activities provide a direct and measurable public-engagement pathway appropriate to the scale and fundamental methodological nature of Q-SHIELD.

Economic and technological impact. Direct commercial impact during the fellowship is expected to be limited because Q-SHIELD develops fundamental statistical methodology. Its technological contribution will arise through reusable inference methods, validation protocols, benchmarking resources, and research software supporting R&D in quantum computing, photonics, communication, and sensing. Experimental quantum groups, research infrastructures, quantum-software developers, and research-software engineers will be able to adapt these outputs for privacy-aware inference, robustness assessment, and reproducible benchmarking. These outputs can reduce methodological barriers arising from uncertain privacy–utility costs, physically inconsistent inference, and the lack of reproducible benchmarking under realistic noise, loss, and finite samples, providing an enabling foundation for subsequent quantum-technology R&D beyond the fellowship.

3. Quality and Efficiency of the Implementation

3.1 Quality and effectiveness of the work plan, assessment of risks and appropriateness of the effort assigned to work packages

Work Plan and Effort. Q-SHIELD follows a staged and overlapping work plan from privacy-preserving statistical foundations (WP1), through structure-aware inference under privacy constraints (WP2), to experimental and computational validation (WP3). The M10–M12 overlap transfers WP1 privacy guarantees and statistical bounds into WP2, while the M19–M21 overlap enables mature WP2 methods to enter early validation. The 24-PM effort is distributed across the three scientific WPs and the transversal management, training and D&E&C activities, as detailed in Table 5.

Table 5. Work plan, effort, deliverables and milestones

WP / Activity	Timing / Effort	Objective & main tasks	Deliverables / Milestones
WP1 – Privacy-Preserving Statistical Estimation	M1–M12 / 7 PM	Establish privacy-preserving finite-sample estimation theory and quantify the statistical cost of privacy. T1.1 Gaussian moment estimation; T1.2 Hamiltonian-parameter estimation; T1.3 privacy–utility guarantees, composition and controlled extension.	D1.1 M6: Privacy-preserving estimation framework. D1.2 M12: Minimax, privacy–utility and composition guarantees. MS1 M12: Privacy foundations established for WP2.
WP2 – Structure-Aware Statistical Inference under Privacy Constraints	M10–M21 / 8 PM	Exploit Hamiltonian and physical structure to retain efficient, physically consistent inference under WP1 privacy constraints. T2.1 Hamiltonian-guided state inference; T2.2 recovery of physical quantities and UQ; T2.3 controlled near-Gaussian extension.	D2.1 M18: Structure-aware state-inference methodology. D2.2 M21: Physically consistent inference with UQ/privacy–utility evaluation. MS2 M21: Framework ready for final validation.
WP3 – Experimental & Computational Validation	M19–M24 / 5 PM	Test privacy–utility advantages and physical reliability under realistic conditions. T3.1 GBS experimental validation at Oxford; T3.2 large-scale benchmarking at QuNorth; T3.3 robustness under noise, loss and finite samples.	D3.1 M21: GBS validation results. D3.2 M24: Benchmarking and robustness assessment. MS3 M24: Validation completed.
Management	M1–M24 / part of 4 PM transversal effort	Scientific coordination, progress and risk monitoring, reporting, data/Open Science management.	DMP M3; progress reviewed at milestones.
Training	M1–M24 / part of 4 PM transversal effort	Scientific and transferable-skills training according to Section 1.3.2; supervision and career development.	CDP M6, reviewed at key milestones; targeted training programme implemented.
D&E&C	M1–M24 / part of 4 PM transversal effort	Dissemination, exploitation, Open Access and public engagement according to Section 2.2.	D&E&C strategy from M1 , updated throughout; formal final plan in penultimate month.
NAP – Research Software Translation	M25–M30 / additional 6 months	Translate validated Q-SHIELD methods into reusable research software and reproducible benchmarking workflows at SIMULA.	NAP-D1 M30: Reusable software and benchmarking package. MS4 M30: NAP completed.

Funding of Research and Training Activities. The MSCA Research, Training and Networking allowance, together with infrastructure and support provided by the participating organisations, is sufficient for the planned theoretical and computational research, training, dissemination, and mobility activities. Access to specialised photonic and computational infrastructure is provided through Oxford and QuNorth, reducing additional infrastructure costs.

Secondments and Non-academic Placement. The Oxford secondment (M19–M21) supports WP3 experimental validation using programmable photonic/GBS platforms. The QuNorth secondment (M22–M24) provides access to the Magne infrastructure for large-scale benchmarking of privacy–utility performance, robustness, and scalability. Their sequential timing enables experimental evidence from Oxford to inform the subsequent large-scale computational assessment. The six-month non-academic placement at SIMULA (M25–M30), following the 24-month European Fellowship, translates the validated methodology into reusable research software and reproducible benchmarking workflows while strengthening software-engineering and intersectoral competences.

	Tasks	M1-M3	M4-M6	M7-M9	M10-M12	M13-M15	M16-M18	M19-M21	M22-M24	M25-M27	M28-M30
WP1	T1.1		D1.1								
	T1.2										
	T1.3				D1.2						
WP2	T2.1										
	T2.2						D2.1				
	T2.3							D2.2			
WP3	T3.1								D3.1		
	T3.2										
	T3.3								D3.2		
NAP											NAP-D
Management	DMP										
Training		UiO						Oxford	QuNorth	SIMULA	
D&E&C			CDP								
Secondment								Oxford	QuNorth		
Placement										SIMULA	
Milestones(MS)					MS1			MS2	MS3		MS4

Figure 2. Gantt Chart

Risk Management. Scientific and administrative risks will be monitored through regular supervision and milestone reviews. The staged Gaussian-to-near-Gaussian strategy limits scientific risk: the analytically tractable Gaussian framework constitutes the core methodology, with controlled near-Gaussian extensions introduced incrementally. Specific risks, likelihood/consequence, and mitigation measures are summarized in Table 6.

Table 6. Risk Management Plan

Risk	Activity	L/C	Mitigation / contingency
Excessive accuracy loss under privacy	WP1	M/M	Adjust privacy mechanism/budget; exploit energy constraints and finite-sample bounds
High computational cost of structure-aware inference	WP2	M/M	Exploit sparsity/locality; use scalable approximations and simpler baselines
Near-Gaussian extension more difficult than expected	WP1–WP2	M/M	Complete and validate Gaussian core; extend incrementally to controlled perturbations
Experimental/HPC access delayed	WP3	L/M	Use simulated/available benchmark data; reschedule external validation
Coordination or mobility delays	Management	L/L	Early planning, regular reviews and remote collaboration where needed
Software/release delays	D&E&C/NAP	L/L	Progressive documentation/releases; prioritise core reproducible outputs

3.2 *Quality and capacity of the host institutions and participating organisations, including hosting arrangements*

Hosting and integration. The fellow will be fully integrated into the Department of Physics and QONTROL at UiO through regular supervision by Professor Morten Hjorth-Jensen, group meetings, seminars and interdisciplinary activities, where he will present project progress, interact with researchers in quantum modelling and computation, and contribute his expertise in statistics and DP. Opportunities for co-publication and co-supervision will further support scientific integration and two-way knowledge transfer. During the secondments, the fellow will be hosted within Professor Ian A. Walmsley's research environment at Oxford (M19–M21) for programmable photonic/GBS validation and at QuNorth (M22–M24) for Magne-based computational benchmarking; the SIMULA NAP (M25–M30), hosted with Dr Shaukat Ali, provides an intersectoral environment for research software engineering and reproducibility.

Institutional support. UiO provides the administrative and practical support required for effective implementation. The Faculty EU Team supports project administration and financial management, while UiO's central EU support covers relevant legal matters. The International Staff Mobility Office supports relocation, residence formalities and practical integration in Norway, and the UiO Library provides Open Science, Open Access and research-data-management support. Career-development services complement the scientific training described in Section 1.3. Successful eligible European PF fellows at the Faculty of Mathematics and Natural Sciences are also offered an internally funded additional year outside the MSCA action, providing further time for exploitation of results, follow-on funding applications and career consolidation.

Quality and capacity. UiO combines Professor Hjorth-Jensen's expertise in Hamiltonian modelling, computational physics and quantum information with the interdisciplinary QONTROL environment and the computing, library and research support required for Q-SHIELD. Oxford adds programmable photonic/GBS capability, QuNorth provides large-scale computational infrastructure through Magne, and SIMULA contributes research software engineering and reproducibility expertise.

References

- [1] Weedbrook et al. (2012). *Rev. Mod. Phys.* 84, 621–669.
- [2] Adesso et al. (2014). *Open Syst. Inf. Dyn.* 21.
- [3] Hamilton et al. (2017). *Phys. Rev. Lett.* 119, 170501.
- [4] Bulmer et al. (2021). *Sci. Adv.* 8.
- [5] Yu et al. (2022). *Nat. Comput. Sci.* 3, 839–848.
- [6] Walschaers (2021). *PRX Quantum*.
- [7] Mele et al. (2024). *Nat. Phys.* 21, 2002–2008.
- [8] Fanizza et al. (2024). *arXiv:2411.03163*.
- [9] Kamath et al. (2019). *IEEE Trans. Inf. Theory (IEEE TIT)*.
- [10] Park et al. (2016). *Proc. AISTATS*.
- [11] Su, (2025) *Annu. Rev. Stat. Appl.* 12, 157–175.
- [12] Dasgupta et al. (2025). *arXiv:2511.01467*.
- [13] Bittel et al. (2025). *arXiv:2508.14979*.
- [14] Cook et al. (2025). *Nat. Commun.* 16, 5929.
- [15] Kumagai et al. (2011). *Commun. Math. Phys.* 318, 535–574.
- [16] Ma et al. (2025). *Natl. Sci. Rev.* 12, nwaf269.
- [17] Oh et al. (2024). *Phys. Rev. Lett.* 133, 230604.
- [18] Dasgupta et al. (2026). *arXiv:2602.01177*
- [19] Caro et al. (2024). *Proc. COLT*.
- [20] Filip et al. (2011). *Phys. Rev. Lett.* 106, 200401.
- [21] Banchi et al. (2015). *Phys. Rev. Lett.* 115, 260501.
- [22] Wu (2025). *Sci. Rep.* 15, 1015.
- [23] Wu (2021). *Sci. Rep.* 11, 11009.
- [24] Wu (2021). *Comput. Intell. Neurosci.* 2021, 7962489.
- [25] Yang et al. (2023). *Sci. Rep.* 13, 5819.
- [26] Wu (2026). *Math. Methods Appl. Sci.* 49, 3628–3638.

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